

Under review as a conference paper at ICAIS 2025

DYNAMIC HYBRID VARIATIONAL-IMPORTANCE
WEIGHTING FOR INCOMPLETE HIGH-DIMENSIONAL
DATA

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Paper under double-blind review

ABSTRACT

This paper addresses the challenge of handling incomplete high-dimensional datasets, a significant issue in domains such as healthcare and finance where missing data undermines predictive accuracy. Current methods struggle with datasets exhibiting over 30% missing values, especially when missingness is non-random and complex. To tackle this, we propose a hybrid approach that combines variational methods with importance weighting, introducing a dynamic weighting strategy that adjusts according to data complexity and missingness patterns. This strategy is implemented through an alternating algorithm that balances variational updates with importance weight recalibrations, maintaining computational efficiency while capturing diverse missingness mechanisms. Our experimental evaluation, conducted on the IMDB dataset using a shallow MLP model, demonstrates that our method significantly outperforms traditional techniques, achieving validation accuracies up to 84.65% with corresponding F1 scores of 0.8505. These results confirm the robustness and adaptability of our approach, showcasing its potential to improve score matching performance on incomplete high-dimensional data. Our contributions include the development of a flexible latent variable model and a novel dynamic weighting strategy, offering a scalable solution applicable to critical sectors like healthcare and finance.

1 INTRODUCTION

The persistent challenge of handling incomplete high-dimensional datasets has gained significant attention in recent years, particularly in critical domains such as healthcare and finance where data integrity is paramount. These sectors rely heavily on robust data-driven decision-making, making the presence of missing data a critical bottleneck. Traditional methods often falter with datasets exhibiting more than 30% missing values, especially under non-random missingness patterns that complicate model training and degrade predictive accuracy. This raises a critical question: Can a hybrid approach that effectively leverages both variational methods and importance weighting improve the performance of score matching in these challenging datasets (Givens et al., 2025)? Addressing the issue of missing data is crucial for the research community, given the increasing reliance on data to inform decisions in sensitive applications. The demand for robust and reliable models that can handle incomplete data is heightened by the need for accurate predictions in areas like clinical diagnostics and financial forecasting, where decision errors can have severe consequences. Recent advances in variational inference have shown promise in managing arbitrary patterns of missing data, including mixed frequencies and sample sizes (Geffner & Donike, 2022), contributing to the urgency of developing more sophisticated solutions. Techniques for imputation and clustering, such as those combining K-Nearest Neighbors (KNN) and DBSCAN, have also been explored to robustly manage incomplete data (Sani et al., 2024). The complexity of this problem is amplified by the intrinsic characteristics of high-dimensional datasets, such as sparsity and multicollinearity, which complicate parameter estimation and model training. Naive approaches, such as mean imputation or complete-case analysis, often fail to address these challenges adequately. Variational Bayesian inference, while offering improvements in quantile regression models dealing with nonignorable missing data, still faces scalability issues when combined

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with importance weighting (Maged et al., 2025). Additionally, the unpredictability of non-random missingness patterns in real-world datasets further complicates model development, rendering many existing solutions inadequate (Duan et al., 2021; Stokes et al., 2025). The use of residual importance

AI Review from DeepReviewer

 AI Review from DeepReviewer will be automatically processed

Summary

This paper introduces a hybrid approach to address the challenge of handling incomplete high-dimensional datasets, a common issue in critical domains like healthcare and finance. The authors propose a method that combines variational methods with importance weighting, incorporating a dynamic weighting strategy that adjusts based on data complexity and missingness patterns. The core idea is to leverage the strengths of both variational inference and importance weighting to improve the robustness and accuracy of models trained on incomplete data. The method is evaluated on the IMDB dataset using a shallow multilayer perceptron (MLP) model, and the results demonstrate improved performance compared to traditional techniques, particularly in scenarios with substantial missing data. The authors present a hybrid score matching objective that blends the variational inference loss and the importance weighting loss using a dynamically adjusted parameter. The variational component employs a deep neural network to approximate intractable distributions, while the importance weighting component recalibrates the model's focus based on observed missingness patterns. The dynamic weighting strategy is a key innovation, allowing the model to adapt to different data complexities and missingness patterns. The experimental results, while limited to a single dataset, show promising performance improvements in terms of validation accuracy and F1 scores. The authors claim that their method outperforms traditional score matching techniques, especially under non-random missingness conditions. The paper's significance lies in its potential to offer a more robust approach to handling incomplete data, which is a pervasive problem in many real-world applications. However, the paper also acknowledges limitations, particularly in scalability to extremely large datasets and handling extreme missingness scenarios. Despite these limitations, the proposed method represents a valuable step towards more effective handling of incomplete high-dimensional data. The authors also suggest future research directions, including further algorithmic optimizations and exploring more advanced strategies for handling extreme missingness. Overall, the paper presents a novel and potentially impactful approach to a significant problem, but it also highlights the need for further research and development to address its limitations and fully realize its potential.

Strengths

The paper's primary strength lies in its innovative approach to handling incomplete high-dimensional data by combining variational methods with importance weighting. This hybrid approach is a creative solution that leverages the strengths of both techniques, potentially offering a more robust and accurate method for dealing with missing data. The introduction of a dynamic weighting strategy is a key innovation, allowing the model to adapt to varying data complexities and missingness patterns. This adaptability is crucial in real-world scenarios where data quality and completeness can vary significantly. The authors' attempt to integrate these two distinct approaches is a novel contribution to the field. The paper also clearly articulates the problem of handling incomplete data, particularly in high-dimensional settings, and the limitations of existing methods. The proposed hybrid approach is presented as a potential solution to these limitations. The experimental results, although limited in scope, demonstrate promising performance improvements compared to traditional techniques, particularly in terms of validation accuracy and F1 scores. This provides initial evidence of the potential effectiveness of the proposed method. The authors also acknowledge the limitations of their method, particularly in scalability and handling extreme missingness, which is a sign of intellectual honesty and a commitment to further research. The paper's discussion of future research directions, including further algorithmic optimizations and exploring more advanced strategies for handling extreme missingness, indicates a clear path forward for future work. The paper's focus on a practical problem with significant real-world implications, particularly in healthcare and finance, adds to its overall significance. The authors' attempt to address the limitations of traditional score matching techniques, especially under non-random missingness conditions, is also a notable contribution. The paper's clear articulation of the problem, the proposed solution, and the experimental results makes it accessible and easy to follow. The use of a formal problem definition and a detailed description of the algorithm helps in understanding the approach. The paper's potential to offer a more robust approach to handling incomplete data, which is a pervasive problem in many real-world applications, is a significant strength.

Weaknesses

After a thorough review of the paper, I have identified several significant weaknesses that warrant careful consideration. The most prominent issue is the lack of a detailed and rigorous explanation of the proposed method, particularly concerning the dynamic weighting strategy. While the paper introduces the hybrid score matching objective and mentions the dynamic adjustment of the parameter α , it fails to provide a clear mathematical formulation of how this adjustment is implemented. The paper states that α is adjusted based on data complexity and missingness patterns, citing several works, but it does not specify the exact mechanism or function used for this adjustment. This lack of clarity makes it difficult to understand the precise mechanisms of the hybrid approach and hinders the reproducibility of the method. The description of the variational component also lacks sufficient detail. While the paper mentions the use of a deep neural network to approximate intractable distributions and provides the Evidence Lower Bound (ELBO) formula, it does not specify the exact architecture of the neural network or the specific distributions used for the latent variables and the decoder. This lack of detail makes it challenging to understand the specific choices made in the implementation of the variational component. The algorithmic steps are presented in Algorithm 1, but they are at a high level and lack detailed implementation specifics. This lack of clarity in the method description is a significant weakness that undermines the paper's overall contribution. My confidence in this assessment is high, as the absence of a mathematical formulation for the dynamic weighting strategy and the lack of specific details about the variational component are clearly evident in the paper. Another significant weakness is the limited scope of the experimental evaluation. The paper evaluates the method on only a single dataset, the IMDB dataset, and using a shallow MLP model. This raises concerns about the generalizability of the findings. The IMDB dataset, while having high dimensionality, might not be representative of the diverse types of incomplete data encountered in real-world applications. The use of a single shallow MLP model also limits the assessment of the method's performance with different model architectures. The paper lacks comparisons with state-of-the-art methods for handling incomplete data. While the Discussion section mentions that traditional baseline methods underperform, no specific baseline methods are explicitly defined or their results presented in the Experiments section. This lack of comparison makes it difficult to assess the true contribution of the proposed approach. The paper also does not explore the method's performance under different missingness mechanisms or varying degrees of missing data. This is a significant omission, as the performance of methods for handling incomplete data can vary significantly depending on the missingness mechanism and the amount of missing data. The absence of experiments with different missingness mechanisms and varying degrees of missing data limits the assessment of the method's robustness. My confidence in this assessment is high, as the use of a single dataset and model, the absence of explicit baseline comparisons, and the lack of experiments with different missingness mechanisms are clearly evident in the paper. Finally, while the paper does touch upon limitations related to scalability and extreme missingness in the Discussion section, it does not delve deeply into the computational cost of the proposed method or provide a detailed analysis of how the method might perform on datasets with different characteristics or under different assumptions beyond the mentioned scalability and missingness issues. The paper lacks a thorough discussion of the computational complexity of the proposed method, particularly in relation to the size of the dataset and

the dimensionality of the input features. It would be beneficial to include a theoretical analysis of the time and space complexity, and to compare it with the complexity of existing methods. Furthermore, the authors should discuss the practical limitations of the method, such as the memory requirements and the potential for numerical instability when dealing with very high-dimensional data. This analysis should also consider the impact of different missingness patterns on the computational cost, as some patterns may be more challenging to handle than others. For example, a discussion on how the method scales with the number of missing values and the percentage of missing data would be valuable. My confidence in this assessment is medium, as the paper does acknowledge some limitations but does not provide a thorough discussion of computational cost and performance on diverse datasets.

Suggestions

To address the identified weaknesses, I recommend several concrete and actionable improvements. First and foremost, the authors should provide a more detailed and rigorous explanation of their proposed method. This should include a clear mathematical formulation of the dynamic weighting strategy, specifying how the importance weights are calculated and adjusted based on data complexity and missingness patterns. The paper should also explain how this dynamic weighting integrates with the variational inference framework. For example, the authors could provide equations that show how the weights are incorporated into the variational loss function. A step-by-step algorithm outlining the entire training process would also be beneficial. Furthermore, the variational component needs a more thorough explanation, detailing the choice of distributions, the approximation techniques used, and the specific form of the Evidence Lower Bound (ELBO) being optimized. The authors should also clarify the specific choices of the variational distributions and the encoder/decoder architectures used in their model. This level of detail is crucial for the reproducibility and understanding of the proposed method. To strengthen the experimental evaluation, the authors should include a more diverse set of datasets, including those with different characteristics and from different domains. It is important to evaluate the method's performance under different missingness mechanisms (e.g., missing completely at random, missing at random, missing not at random) and varying degrees of missing data. The paper should also include comparisons with state-of-the-art methods for handling incomplete data, such as imputation-based approaches or other hybrid methods. The evaluation should not be limited to a single shallow MLP model; the authors should explore the performance of the proposed method with different model architectures and hyperparameter settings. The results should be presented with appropriate statistical analysis, including confidence intervals and significance tests, to demonstrate the robustness of the findings. Additionally, the authors should provide a detailed analysis of the computational cost of the proposed method and its scalability to larger datasets. This should include a theoretical analysis of the time and space complexity, and a comparison with the complexity of existing methods. The authors should also discuss the practical limitations of the method, such as the memory requirements and the potential for numerical instability when dealing with very high-dimensional data. This analysis should also consider the impact of different missingness patterns on the computational cost. Furthermore, the authors should provide a more in-depth comparison with existing methods for handling incomplete high-dimensional data. This comparison should not only focus on the performance of the methods, but also on their underlying assumptions, limitations, and computational complexity. The authors should discuss the specific advantages and disadvantages of their proposed method compared to existing approaches, and highlight the scenarios where their method is expected to perform best. This comparison should also include a discussion of the theoretical underpinnings of the different methods, and how they relate to the specific problem of handling incomplete high-dimensional data. The authors should also discuss the potential for combining their method with other techniques, such as imputation or data augmentation, to further improve performance. Finally, the authors should provide a more thorough discussion of the limitations of the proposed method and potential areas for future research. This should include a discussion of the assumptions underlying the method and how these assumptions might affect its performance. The authors should also explore the sensitivity of the method to different hyperparameter settings and provide guidelines for choosing appropriate values. Potential areas for future research could include extending the method to handle different types of data, such as time-series data or categorical data, and exploring the use of more advanced variational inference techniques. The authors should also consider the ethical implications of their work, particularly in the context of high-stakes domains like healthcare and finance.

Questions

After reviewing the paper, I have several questions that I believe are crucial for a deeper understanding of the proposed method and its implications. First, how does the proposed dynamic weighting strategy specifically adjust to different patterns of missingness in the data? Can you provide a more detailed explanation or examples? The paper mentions that the parameter α is dynamically adjusted based on data complexity and missingness patterns, but it does not provide the exact mechanism or function used for this adjustment. Understanding this mechanism is crucial for evaluating the effectiveness of the proposed method. Second, what are the computational costs associated with your method, and how does it compare to traditional approaches in terms of efficiency? The paper does not provide a detailed analysis of the computational complexity of the proposed method, particularly in relation to the size of the dataset and the dimensionality of the input features. A thorough analysis of the time and space complexity, and a comparison with the complexity of existing methods, would be valuable. Third, how does your method perform on datasets with different characteristics or from different domains? Have you considered evaluating it on a more diverse set of datasets? The experimental evaluation is limited to a single dataset (IMDb), and it is unclear how the method would perform on datasets with different characteristics or from different domains. Evaluating the method on a more diverse set of datasets would help to demonstrate its robustness and generalizability. Fourth, can you provide a more detailed comparison of your method with existing state-of-the-art approaches for handling incomplete high-dimensional data? What are the specific advantages of your method over these existing approaches? The paper mentions some related work, but it does not provide a detailed comparative analysis of specific methods. A more in-depth comparison with existing methods, including their underlying assumptions, limitations, and computational complexity, would be valuable. Fifth, how does the proposed method scale to extremely large datasets with very high-dimensional features? Are there any computational bottlenecks or limitations that need to be addressed? The paper acknowledges that scalability is a limitation, but it does not provide a detailed analysis of the computational bottlenecks or limitations that need to be addressed. Sixth, how does the method handle datasets with very high missingness rates (e.g., more than 50%)? Are there any specific strategies or adjustments needed for such scenarios? The paper does not explore the method's performance under different degrees of missing data, and it is unclear how the method would handle datasets with very high missingness rates. Finally, the paper mentions that the dynamic weighting strategy is a key innovation. Can you provide more details on how this strategy is implemented and how it adapts to different data complexities and missingness patterns? The paper does not provide a clear mathematical formulation of the dynamic weighting strategy, and more details on its implementation would be beneficial.

Scores

Soundness: **1.5**

Presentation: **1.5**

Contribution: **1.5**

Rating: 2.5



Summary

The paper proposes a hybrid learning objective for handling incomplete high-dimensional data that linearly combines a variational component and an importance-weighted component with a dynamically adjusted mixing coefficient α . Training alternates between variational updates, recalibration of importance weights, and dynamic adjustment of α (Algorithm 1). The method is positioned as improving "score matching" under complex missingness, and is evaluated on IMDb (9,000 samples, 1,000-dim TF-IDF) using a shallow MLP. Reported validation accuracies are up to 84.65% with F1 around 0.85 over three runs. The authors claim superior performance over traditional methods and emphasize scalability and robustness to non-random missingness.



Strengths

- Addresses an important problem: learning from high-dimensional data with substantial missingness, potentially non-random (Sections 1, 3).
- Conceptual proposal of a hybrid objective that balances variational inference and importance weighting with a dynamic α (Eq. (1), Algorithm 1).
- Clear, simple experimental setup (IMDb TF-IDF; shallow MLP) that is easy to scale and potentially reproduce at a high level (Section 4).
- Multiple runs reported with accuracy and F1 metrics (Table 1), indicating at least some attention to repeatability.



Weaknesses

- Methodological underspecification: the paper claims a score matching formulation, but the loss components are not defined as score-matching objectives. In particular, Eq. (4) is a weighted log-likelihood term, and Implementation Details state training uses binary cross-entropy for classification (Section 4), creating a mismatch with the claimed contribution to score matching.
- Dynamic weighting is a central contribution, but the adjustment mechanism for α is not specified (what inputs, update rule, schedule, controller, or criteria?), preventing reproducibility and hindering verification of the claimed benefits (Algorithm 1; Section 3).
- Importance weights w_i are undefined beyond a high-level description (no derivation/estimation procedure, propensity model, density ratio estimator, normalization/clipping, variance control, or effective sample size diagnostics), which is crucial for stability and bias-variance trade-offs (Eq. (4), Section 3).
- Missing data generation and mechanism are not described: no missingness rate, no MCAR/MAR/MNAR specification, no mask modeling, and no details on how m is used in the encoder/decoder or classifier (Sections 3–4).
- Comparative evaluation is incomplete: the paper claims to outperform "traditional techniques" but does not name or define baselines, give their setups/hyperparameters, or report statistical significance (Sections 1, 5; Abstract).
- Limited experimental scope: a single dataset (IMDb TF-IDF), 5 training epochs, no ablations (e.g., static vs dynamic α , VI-only vs IW-only), no sensitivity to missingness level/mechanism, and no wall-clock or memory cost (Sections 4–5).
- Algorithmic clarity issues: Algorithm 1 is a high-level loop without pseudo-code for the key steps; the final line "end for=0" appears erroneous; the latent variable model (Eqs. (2)–(3)) is not integrated with the shallow MLP classifier in implementation (Sections 3–4).



Questions

- Score matching vs. likelihood: What are the exact mathematical forms of L_{var} and L_{imp} when the objective is "score matching"? As written, Eq. (4) is a weighted log-likelihood, and Section 4 uses binary cross-entropy. Please reconcile the score-matching claim with the implemented loss and provide explicit formulas.
- Dynamic α : How is α updated? What signals (e.g., validation loss, gradient variance, missingness statistics, ESS) drive the adjustment? Provide the exact update rule/schedule with hyperparameters, and an ablation comparing static vs dynamic α .
- Importance weights: How are w_i computed in practice under missingness? Are they propensity scores, density ratios, or residual weights? How are they estimated, normalized/clipped, and regularized to control variance? Report effective sample size and stability diagnostics.
- Missingness: What missingness mechanisms and rates are used (MCAR/MAR/MNAR)? How are masks m generated or observed in IMDb, and how are they incorporated into $q(z|x,m)$ and the classifier? Provide results stratified by missingness level and mechanism.
- Baselines: Which "traditional techniques" were compared? Please list and detail their configurations (e.g., MICE/MissForest/GAIN/MIWAE/Not-MIWAE, VI-only, IW-only, standard score matching without weighting), and report fair hyperparameter tuning and statistical tests.
- Architecture integration: How is the latent variable model (Eqs. (2)–(3)) implemented alongside the shallow MLP? Is the VAE used end-to-end with the classifier, or are embeddings precomputed? Provide a schematic and pseudo-code.
- Optimization details: What are batch size, number of importance samples (if any), learning rate schedules, early stopping criteria, and number of epochs needed for convergence? Provide training time and memory compared to baselines.
- Generalization: Have you evaluated on additional datasets (tabular and text) and under domain shift? How does the method perform beyond IMDb?
- Reproducibility: You state code is provided. Please include a link (in the camera-ready), commit hash, and scripts to reproduce all tables/figures with fixed seeds.

 Limitations

- Methodological ambiguity limits reproducibility: unspecified alpha update and w_i estimation could lead to irreproducible gains and brittle behavior.
- Potential high-variance estimation from importance weighting under high-dimensional missingness; without stabilization (e.g., clipping, control variates), training may be unstable.
- Identifiability concerns under MNAR without explicit missingness modeling or sensitivity analysis; dynamic weighting may not remedy fundamental non-identifiability.
- Scalability beyond IMDB and shallow models is untested; alternating VI/IW updates may be costly for large models and datasets without careful engineering.
- Possible societal impacts: In healthcare/finance, improperly estimated weights or dynamics could exacerbate biases if missingness correlates with protected attributes; overconfident predictions on imputed patterns could mislead downstream decisions. Mitigations include fairness-aware weighting, sensitivity analyses, and uncertainty reporting.

 Image Evaluation

Cross-Modal Consistency: 16/50

Textual Logical Soundness: 10/30

Visual Aesthetics & Clarity: 12/20

Overall Score: 38/100

Detailed Evaluation (≤ 500 words):

1. Cross-Modal Consistency

- Major 1: Core “score matching” claim conflicts with a supervised BCE setup; no score objective/metrics are shown. Evidence: Sec 3 Eq.(1) vs Sec 4 “binary cross-entropy loss function”.
- Major 2: L_{imp} uses generative $\log p(x|\theta)$ while the task is labeled IMDB sentiment classification (y never defined). Evidence: Eq.(4) “ $\log p(x_i | \theta)$ ” vs Sec 4 “binary classification task”.
- Major 3: Dynamic weight α touted as key but lacks a definition, rule, or features for “data complexity”. Evidence: Sec 3 “ α is dynamically adjusted based on data complexity” and Alg. 1 “Adjust α ...”.
- Major 4: Variational notation is inconsistent with VAEs ($p(x|z) \approx q(x|z;\varphi)$; $p(z|x,m) \approx q(\dots)$). Evidence: Eq.(2) “ $p(x|z) \approx q(x|z; \varphi)$ ” and Eq.(3).
- Major 5: Claims superior baselines without any comparative figure/table, preventing verification. Evidence: Sec 5 “Traditional baseline methods... underperform” vs only Table 1.
- Minor 1: Algorithm block formatting error suggests editing issues. Evidence: Alg. 1 “end for=0”.
- Minor 2: Table 1 shows identical validation and test scores in all runs, suggesting leakage or unclear protocol. Evidence: Table 1 (val = test for runs 1–3).

2. Text Logic

- Major 1: Central novelty (hybrid variational–importance weighting for score matching) lacks an operational definition for α and w_i and any score-matching formulation. Evidence: Sec 3 Eqs.(1–4) omit α rule and w_i computation.
- Major 2: Missingness mechanism and protocol (rate, MAR/MNAR, mask generation, use in training) not specified though it is the core setting. Evidence: Sec 4 Dataset/Training contain no missingness description.
- Minor 1: Baseline comparison is asserted but not supported by numbers, ablations, or statistical tests. Evidence: Sec 5 narrative only.
- Minor 2: Notation flip between p and q for generative/variational distributions undermines clarity. Evidence: Sec 3 Eqs.(2–3).

3. Figure Quality

- Major 1: Table 1 omits baselines, variance/error bars, and protocol notes; identical val/test obscures the intended message. Evidence: Table 1 contents.
- Minor 1: No figure/plot of α schedule or w_i distribution; readers cannot grasp the “dynamic” component. Evidence: No figures aside from Table 1.
- Minor 2: Algorithm 1 is too high-level to stand alone (no α update rule, no w_i formula). Evidence: Alg. 1 step list only.

Key strengths:

- Clear dataset size and simple MLP architecture aid reproducibility.
- Concise results table with multiple runs; reported numbers internally consistent.

Key weaknesses:

- Method–experiment mismatch (score matching vs supervised BCE).
- Undefined α and w_i ; missingness protocol unspecified.
- No baseline/table for claimed superiority; no ablations or variance.
- Inconsistent variational notation; generative objective misaligned with task.

 Scores

Originality: 2	Quality: 2	Clarity: 2	Significance: 2	Soundness: 2
Presentation: 2	Contribution: 2			

Rating: 4

AI Review from SafeReviewer

AI Review from SafeReviewer will be automatically processed

Summary

This paper introduces a hybrid approach for handling incomplete high-dimensional datasets, combining variational methods with importance weighting. The core idea is to dynamically adjust the balance between these two techniques based on the data's complexity and the observed patterns of missingness. The authors propose a novel dynamic weighting strategy, implemented through an alternating algorithm that performs variational updates and importance weight recalibrations. The variational component leverages a latent variable model, while the importance weighting component aims to correct for biases introduced by missing data. The authors evaluate their method on the IMDB dataset, a text dataset, using a shallow multilayer perceptron (MLP) model. The primary goal of the experiment is to demonstrate the effectiveness of the proposed hybrid approach in improving predictive performance compared to not explicitly addressing missing data. The results, presented as validation and test accuracies and F1 scores, are reported to be consistent across multiple runs, suggesting some level of robustness. The authors position their work as particularly relevant for domains like healthcare and finance, where missing data is a common challenge. However, the paper lacks a thorough comparison with existing state-of-the-art methods for handling missing data, and the experimental evaluation is limited to a single dataset and model architecture. While the core idea of combining variational inference and importance weighting is interesting, the paper needs further development to demonstrate its practical utility and generalizability. The lack of detailed methodological explanations and the absence of a comprehensive experimental evaluation limit the impact of the work. The paper presents a potentially valuable approach, but its current form requires significant improvements to be considered a robust contribution to the field.

Strengths

The paper's primary strength lies in its clear articulation of the problem it seeks to address: the challenge of handling incomplete high-dimensional datasets, a common issue in many real-world applications. The authors correctly identify the limitations of traditional methods in dealing with complex missingness patterns and high dimensionality. The motivation for combining variational methods and importance weighting is also well-founded. Variational methods offer a way to model complex data distributions, while importance weighting can correct for biases introduced by missing data. The idea of dynamically adjusting the balance between these two techniques based on data complexity and missingness patterns is a novel and potentially valuable contribution. The authors also present a clear, high-level overview of their proposed method, outlining the variational and importance weighting components and the alternating optimization algorithm. The use of a latent variable model within the variational framework is a standard and appropriate approach for handling complex data distributions. The paper also makes an effort to make the experimental setup reproducible by mentioning the use of PyTorch and providing details about the model architecture and training parameters. The consistent performance across multiple runs, as reported in the results, suggests some level of robustness in the proposed method. The authors also clearly state the problem they are trying to solve and the approach they are taking, which makes the paper easy to follow. The paper also attempts to position its work within the broader context of missing data handling, citing relevant works in variational inference and importance weighting. The authors also make an effort to make the experimental setup reproducible by mentioning the use of PyTorch and providing details about the model architecture and training parameters. The consistent performance across multiple runs, as reported in the results, suggests some level of robustness in the proposed method.

Weaknesses

After a thorough examination of the paper, I've identified several significant weaknesses that undermine its overall contribution. First and foremost, the paper suffers from a critical lack of methodological detail. While the authors provide a high-level overview of their approach, they fail to provide sufficient specifics regarding the implementation of key components. For instance, the dynamic weighting strategy, which is presented as a core innovation, lacks a concrete mathematical formulation. The paper describes the dynamic adjustment of the weight based on data complexity and missingness patterns, but it does not specify the exact function or algorithm used for this adjustment. This lack of detail makes it impossible to understand how the dynamic weighting strategy is implemented and to assess its effectiveness. The paper also lacks specific details about the neural network architectures used for the encoder and decoder in the variational component. While the experimental section describes the MLP used for classification, it does not provide details about the networks used within the variational framework. This lack of information makes it difficult to reproduce the results and to understand the specific choices made by the authors. The paper also fails to provide a detailed explanation of how the importance weights are calculated. While the paper mentions that the importance weights reflect the likelihood of each data point, it does not provide the specific formula or algorithm used for their computation. This lack of transparency makes it difficult to assess the validity of the importance weighting component. My confidence in this weakness is high, as the absence of these details is clearly evident from the paper's content. Second, the paper's experimental evaluation is severely limited. The authors evaluate their method on only a single dataset, the IMDB dataset, which is a text dataset. This narrow evaluation makes it impossible to assess the generalizability of the proposed method to other types of data and missingness patterns. The paper also lacks a direct comparison with established baseline methods for handling missing data. The authors mention that their method outperforms traditional score matching techniques, but they do not provide any experimental results to support this claim. The absence of comparisons with methods like mean imputation, k-Nearest Neighbors (KNN) imputation, or matrix factorization techniques makes it difficult to assess the relative performance of the proposed method. Furthermore, the paper does not include ablation studies to demonstrate the individual contributions of the variational inference and importance weighting components. This lack of ablation studies makes it difficult to understand the specific benefits of combining these two techniques. The paper also does not explore the performance of the proposed method under different missingness mechanisms (MCAR, MAR, MNAR), which is crucial for understanding its robustness. The limited experimental evaluation significantly reduces the impact of the paper. My confidence in this weakness is high, as the lack of diverse datasets, baselines, and ablation studies is clearly evident from the paper's content. Finally, the paper suffers from a lack of relevant literature citations. The paper does not adequately discuss or cite relevant works on missing data imputation, score matching, and variational completion. The absence of these citations makes it difficult to

position the proposed method within the broader context of missing data handling. The paper also lacks a discussion of the limitations of the proposed method and potential avenues for future research. This omission makes the paper less comprehensive and less insightful. My confidence in this weakness is high, as the absence of these citations is clearly evident from the paper's content. The combination of these weaknesses significantly undermines the paper's contribution and makes it difficult to assess the practical utility and generalizability of the proposed method.

 Suggestions

To address the identified weaknesses, I recommend several concrete improvements. First, the authors must provide a detailed and rigorous explanation of their methodology. This includes providing a specific mathematical formulation for the dynamic weighting strategy, detailing how the weight α is adjusted based on data complexity and missingness patterns. The authors should also provide specific details about the neural network architectures used for the encoder and decoder in the variational component, including the number of layers, the activation functions, and the number of parameters. Furthermore, the authors should provide a detailed explanation of how the importance weights are calculated, including the specific formula or algorithm used for their computation. This additional detail will significantly improve the transparency and reproducibility of the work. Second, the authors must significantly expand their experimental evaluation. This includes evaluating their method on multiple datasets, including datasets with different types of data (e.g., images, time series) and different missingness patterns. The authors should also include direct comparisons with established baseline methods for handling missing data, such as mean imputation, KNN imputation, matrix factorization techniques, and other relevant methods. The authors should also conduct ablation studies to demonstrate the individual contributions of the variational inference and importance weighting components. This will help to understand the specific benefits of combining these two techniques. Furthermore, the authors should explore the performance of the proposed method under different missingness mechanisms (MCAR, MAR, MNAR) to assess its robustness. The experimental evaluation should also include a wider range of evaluation metrics, such as precision, recall, and AUC. These changes will significantly improve the validity and generalizability of the results. Third, the authors must include a comprehensive discussion of relevant literature on missing data imputation, score matching, and variational completion. This includes citing and discussing relevant works on these topics and positioning their proposed method within the broader context of missing data handling. The authors should also discuss the limitations of their proposed method and potential avenues for future research. This will make the paper more comprehensive and insightful. Finally, the authors should consider releasing their code to facilitate reproducibility and further research. This will allow other researchers to build upon their work and to validate their findings. By addressing these points, the authors can significantly improve the quality and impact of their work. These suggestions are directly tied to the weaknesses I've identified, and their implementation would substantially strengthen the paper's contribution.

 Questions

After reviewing the paper, I have several questions that I believe are crucial for a deeper understanding of the proposed method and its potential impact. First, regarding the dynamic weighting strategy, I'm curious about the specific criteria used to determine data complexity and missingness patterns. What quantitative measures are used to assess these factors, and how are these measures used to adjust the weight α ? Specifically, is there a specific function or algorithm that governs this adjustment, and if so, what is it? Second, concerning the variational inference component, I'd like to understand the specific architecture of the encoder and decoder networks. What is the number of layers, the activation functions, and the number of parameters in these networks? How were these architectural choices made, and what is the rationale behind them? Third, regarding the importance weighting component, I'm interested in the specific formula or algorithm used to calculate the importance weights. How are these weights computed, and what assumptions are made in this process? Are these weights normalized, and if so, how? Fourth, regarding the experimental evaluation, I'm curious about the specific missingness mechanism used in the IMDb dataset. Was the missing data simulated, and if so, what parameters were used? How does the performance of the proposed method vary under different missingness mechanisms (MCAR, MAR, MNAR)? Fifth, regarding the comparison with baseline methods, I'd like to know why the authors chose not to compare their method with standard imputation techniques like mean imputation, KNN imputation, or matrix factorization. What are the specific reasons for this choice, and what are the authors' expectations regarding the performance of these baseline methods compared to their proposed approach? Sixth, regarding the generalizability of the method, I'm curious about the performance of the proposed method on other types of data, such as images or time series. How does the method perform on datasets with different characteristics and missingness patterns? Finally, regarding the limitations of the method, I'd like to know what the authors consider to be the main limitations of their approach and what are the potential avenues for future research? These questions are designed to address the key uncertainties and methodological choices made by the authors and to seek clarification of critical assumptions and experimental details.

 Scores

Soundness: **1.5**

Confidence: **4.25**

Rating: 2.5

Keywords

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